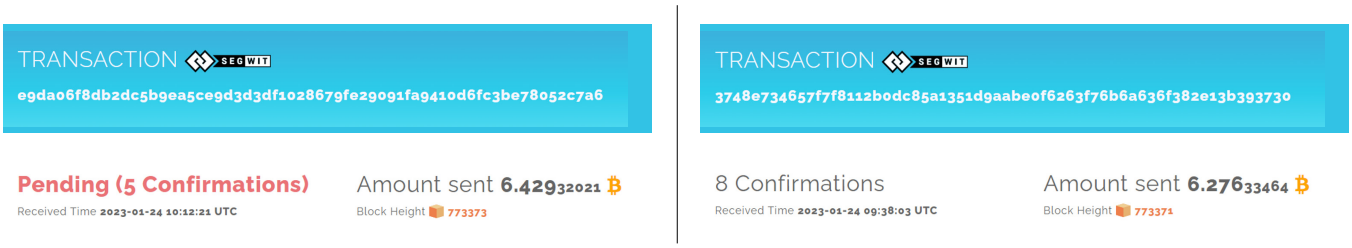
8.5.3 Don't Trust, Verify

In the world of cryptocurrencies, the phrase "**Don't trust, verify**" is a reminder to always verify transactions yourself rather than relying on others, such as a centralized authority or intermediary. The Bitcoin network is made up of a decentralized network of nodes, which allows users to verify transactions on their own.

However, there are scenarios that can cause transactions to be **reversed**, such as **double-spending**, **orphaned blocks**, and **reorganization**. To increase the security of transactions, it is recommended to wait for 6 confirmations, or 6 blocks containing the specific transaction, before considering it final. The more confirmations a transaction has, the more secure it becomes as the likelihood of it being reversed decreases. The number of confirmations needed may vary depending on the use case and desired level of security.

- **Double-spending**: In a double-spend attack, a malicious actor attempts to spend the same bitcoin twice by manipulating the network to accept their second spending of the same ***bitcoin*** as valid. If a miner or group of miners controlling more than 50% of the network's **hashing power** (known as 51% attack) confirm a double-spend transaction, it could be added to a block and considered valid, effectively reversing the original transaction.

- **Orphaned blocks**: When two miners find a new block at the same time, the network may temporarily accept both. When one of the blocks is later extended by additional blocks, the network will recognize this chain as the main chain and the other block will become an orphan, no longer part of the main *blockchain*. The transactions included in the orphaned block are not lost and will be included in a later block if they remain valid.

An orphaned block in ***Bitcoin*** is a valid block that is not included in the longest chain, which is considered the main chain.

- **Reorganization**: This could theoretically happen if a new block were added to the blockchain and it causes the existing chain to be replaced by a different one. If a transaction was included in a block that is no longer on the main chain, it would be considered invalid and the transaction would be reversed.